

Here are the detailed, standalone notes for the video "How to calculate p-values."

Video: How to calculate p-values

Link: <https://youtu.be/JQc3yx0-Q9E>

This video explains the *method* for calculating p-values in two different scenarios: for discrete data (like coin flips) and for continuous data (like height measurements).

Prerequisite: This guide assumes you already know *what* a p-value is and how to interpret it (i.e., that a small p-value is evidence *against* a null hypothesis).

Key Focus: The video focuses on calculating **two-sided p-values**, which are the most common and scientifically robust. It warns that **one-sided p-values** are rarely used and can be misleading.

1. The Core Concept: The Null Hypothesis (H_0)

Before you can calculate a p-value, you must establish your **Null Hypothesis (H_0)**. This is the "default" or "skeptical" position that assumes there is **no effect** or **no difference**.

- **Your Observation:** "I flipped a coin and got two heads in a row. This seems special!" [01:01]
- **Your (Informal) Hypothesis:** "My coin is special."
- **The Null Hypothesis (H_0):** "My coin is **not special**. It's just a normal, fair coin, and getting two heads was just random chance." [01:28]

The p-value calculation is what tests this H_0 . It answers the question: "If the null hypothesis is true (i.e., the coin is normal), what's the probability of getting my observation *or something even more extreme*?"

2. How to Calculate p-values for Discrete Data (e.g., Coin Flips)

For discrete data, we can list all possible outcomes. The p-value is calculated by summing three specific probabilities.

p-value = (Part 1) + (Part 2) + (Part 3)

- **Part 1:** The probability of your specific observation.
- **Part 2:** The probability of all *other* outcomes that are **equally rare** as your observation.
- **Part 3:** The probability of all outcomes that are **rarer (or more extreme)** than your observation.

Example 1: Flipping a coin 2 times and getting 2 Heads (HH)

- H_0 : The coin is normal (50/50 probability).
- **All Possible Outcomes (4 total):**
 - i. **HH** (Heads, Heads)
 - ii. **HT** (Heads, Tails)
 - iii. **TH** (Tails, Heads)
 - iv. **TT** (Tails, Tails)
- **Probabilities of each outcome:**

- $\text{Prob}(\text{HH}) = 1/4 = 0.25$
- $\text{Prob}(\text{TT}) = 1/4 = 0.25$
- $\text{Prob}(\text{one H, one T}) = (\text{HT or TH}) = 2/4 = 0.5$ [03:54]
- **Calculate the p-value for observing HH:**
 - **Part 1 (Observation):** $\text{Prob}(\text{HH}) = 0.25$ [05:02]
 - **Part 2 (Equally Rare):** $\text{Prob}(\text{TT})$ is equally rare as $\text{Prob}(\text{HH})$. So, we add **0.25** [05:11].
 - **Part 3 (Rarer):** Are there any outcomes rarer than HH or TT? No. So, this is **0** [05:25].
 - **Total p-value:** $0.25 + 0.25 + 0 = 0.5$ [05:43]
- **Conclusion:** The p-value is 0.5. Since 0.5 is > 0.05 (the standard cutoff), we **fail to reject the null hypothesis**. We do not have enough evidence to say the coin is special.

Example 2: Flipping a coin 5 times and getting 4 Heads, 1 Tail

- H_0 : The coin is normal.
- **All Possible Outcomes:** 32 total [10:50].
- **Probabilities of all outcome "types":**
 - 5 Heads: 1 way (HHHHH) $\rightarrow \text{Prob} = 1/32$
 - **4 Heads, 1 Tail:** 5 ways (THHHH, HTHHH, etc.) $\rightarrow \text{Prob} = 5/32$
 - 3 Heads, 2 Tails: 10 ways $\rightarrow \text{Prob} = 10/32$
 - 2 Heads, 3 Tails: 10 ways $\rightarrow \text{Prob} = 10/32$
 - **1 Head, 4 Tails:** 5 ways $\rightarrow \text{Prob} = 5/32$
 - 5 Tails: 1 way (TTTTT) $\rightarrow \text{Prob} = 1/32$
- **Calculate the p-value for observing 4 Heads, 1 Tail:**
 - **Part 1 (Observation):** $\text{Prob}(4\text{H}, 1\text{T}) = 5/32$ [11:09]
 - **Part 2 (Equally Rare):** $\text{Prob}(1\text{H}, 4\text{T})$ is equally rare. So, we add **5/32** [11:21].
 - **Part 3 (Rarer):** $\text{Prob}(5\text{H})$ and $\text{Prob}(5\text{T})$ are both rarer ($1/32 < 5/32$). So, we add $(1/32 + 1/32) = 2/32$ [11:40].
 - **Total p-value:** $5/32 + 5/32 + 2/32 = 12/32 = 0.375$ [11:50]
- **Conclusion:** The p-value is 0.375. Since 0.375 is > 0.05 , we **fail to reject the null hypothesis**. This result is not statistically significant.

3. How to Calculate p-values for Continuous Data (e.g., Height)

For continuous data, we can't list all outcomes. We use a **statistical distribution** (a curve), where the **area under the curve** represents probability [12:49].

For a two-sided p-value, we find the probability of our observation *and* the probability of an equally extreme observation on the *other side* of the average.

Example: Distribution of Women's Heights

- H_0 : A person we measure comes from this known distribution.
- **Distribution details:** Average = 155.7 cm. 95% of people are between 142 cm and 169 cm. This means 2.5% are < 142 cm and 2.5% are > 169 cm [13:56].

Observation 1: A person is 142 cm tall.

- The "extremeness" of this point is defined by the area in the tail *beyond* it.
- **Area 1 (Left Tail):** $\text{Prob}(\text{height is } \leq 142 \text{ cm}) = 0.025$ (or 2.5%) [15:30].

- **Area 2 (Right Tail):** We look for the equally extreme point on the *other* side. This is 169 cm. $\text{Prob}(\text{height is } \geq 169 \text{ cm}) = \mathbf{0.025}$ (or 2.5%) [15:55].
- **Total p-value:** $0.025 + 0.025 = \mathbf{0.05}$ [16:15]
- **Conclusion:** The p-value is 0.05. This is exactly on the typical borderline, so the data is inconclusive [16:43].

Observation 2: A person is 141 cm tall.

- This is *more extreme* than 142 cm.
- **Area 1 (Left Tail):** Let's say $\text{Prob}(\text{height} \leq 141 \text{ cm}) = \mathbf{0.016}$ (or 1.6%) [17:07].
- **Area 2 (Right Tail):** The equally extreme point on the other side would also have an area of **0.016** (or 1.6%) beyond it.
- **Total p-value:** $0.016 + 0.016 = \mathbf{0.032}$ [17:15]
- **Conclusion:** The p-value is 0.032. Since 0.032 is < 0.05 , we **reject the null hypothesis**. It is statistically significant. This person is unlikely to have come from this distribution [17:35].

4. The Danger: Two-Sided vs. One-Sided p-values

- **Two-Sided (Standard):** This is what we've been calculating. It looks for a difference in *both* directions (e.g., a drug makes you much *better* OR much *worse*). It will catch any unusual result [23:05].
- **One-Sided (Risky):** Here, you decide *before* the experiment to *only* look in one direction (e.g., "I only care if the drug makes recovery *shorter*") [21:29].
- **The Problem:** What if the drug is harmful and makes recovery *longer*?
 - A **two-sided** test would catch this. It would see the result is extreme (just in the "worse" direction) and give a small p-value [22:57].
 - A **one-sided** test (looking for *shorter* recovery) would completely miss it. The p-value would be huge (e.g., 0.98), and you would incorrectly conclude the drug had no effect, failing to see the harm [23:46].

Final Takeaway: Always use **two-sided p-values** unless you are an expert and have a very specific, justifiable reason not to. They are safer and more honest as they detect any significant difference, whether it's good or bad.

http://googleusercontent.com/youtube_content/7